

1(a)(i)	<i>independent: <u>concentration</u> of calcium chloride / CaCl_2 ;</i> <i>dependent: number of stomata closed / open ;</i>	2	A closing / opening <i>for closed / open</i> I percentage
1(a)(ii)	serial dilution ;	1	A description I simple / standard dilution, or description of I proportional dilution
1(b)(i)	<i>idea of the higher the concentration (of, calcium chloride / CaCl_2,) the greater the, number / percentage / proportion, of stomata that are closed / ora ;</i>	1	<i>hypothesis must be testable and not repeat information given in question</i> A <i>idea that</i> , the number / proportion / percentage of closed stomata is (directly) proportional to the conc. of CaCl_2 A as CaCl_2 concentration increases more stomata close ora A a null hypothesis: different / changing concentrations of CaCl_2 have no (significant) effect on the number / proportion / percentage of, closed / open, stomata

Question	Answer	Mark	Additional Guidance
1(b)(ii)	<i>five from</i> 1 <i>ref. to putting (epidermal) strip(s) in the (different) solutions in appropriate containers ;</i> 2 <i>ref. to keeping in the light (for the investigation) ;</i> 3 <i>ref. to using a (light) microscope (to observe the stomata) ;</i> 4 <i>count/record, (the number of/how many), closed/open stomata ;</i> 5 <i>ref. to standardising the counting ;</i> 6 <i>ref. to making several counts on at least one epidermal strip and taking a mean/to identify anomalies ;</i> 7 <i>max 2 for control variables (mps 7–9)</i> 8 <i>ref. to using suitable equipment for cutting and measuring strips (to same size) ;</i> 9 <i>ref. to method achieving constant temperature ;</i> 10 <i>ref. to method of preventing evaporation ;</i> 11 <i>one of</i> <i>ref. to low risk ;</i> <i>allergy to leaves/plants and wearing gloves/goggles ;</i> <i>CaCl₂ irritant and avoid swallowing/wearing gloves/goggles ;</i> <i>care when cutting with scalpel and cut on tile and away from, hand/body ;</i>	5	A named solutions A e.g. beakers, watch glasses, Petri dishes, test tubes, boiling tubes, measuring cylinders, (microscope) slide/cavity slide I <i>ref. to</i> volume of solution I <i>ref. to</i> time A in dark room with fixed light R electron / electronic microscope I calculate / observe if a number of counts is given it must be a minimum of 3 I average A mean average I repeat / replicate, the experiment <i>unqualified</i> e.g. scalpel or scissors and ruler / calipers I metre ruler e.g. incubator, temperature controlled room, water bath to keep temperature constant e.g. lid / film / coverslip (if slide) AW R no risk I allergy to CaCl₂ I scissors

Question	Answer	Mark	Additional Guidance
1(c)(i)	<i>two (for 1 mark) from</i> (same calibrated eyepiece) graticule used ; (same) microscope ; (same) magnification ;	1	A same calibration for measuring I stage micrometer I same apparatus / method of measuring I random selection of stomata
1(c)(ii)	$0.75 / 7.5 \times 10^{-1} (\mu\text{m}) ;$	1	I $\frac{3}{4}$

Question	Answer	Mark	Additional Guidance
1(d)(i)	<i>one from</i> - up to/at, $0.001 \mu\text{mol dm}^{-3}$ ABA/initially/at first, upper epidermis mean has increased/not changed, lower epidermis has decreased; - lower epidermis responds at $0.001 \mu\text{mol dm}^{-3}$ ABA, upper epidermis responds at $0.01 \mu\text{mol dm}^{-3}$ ABA; - confidence intervals/error bars, do not overlap (until $1.00 \mu\text{mol dm}^{-3}$ ABA); - stomata on upper epidermis have wider aperture at, all/increasing, concentrations of ABA (until $1.00 \mu\text{mol dm}^{-3}$ ABA);	1	<i>idea that</i> upper epidermis at $0.001 \mu\text{mol dm}^{-3}$ has not decreased while lower epidermis has decreased lower epidermis (starts to) responds at lower concentrations of ABA; I standard deviation/standard error I <i>ref. to</i> one stated ABA concentration I <i>ref. to</i> one stated ABA concentration I longer/shorter/higher, aperture/stomata A longer/shorter, diameter/width
1(d)(ii)	<i>one from</i> <i>definition:</i> e.g. the confidence limits are, the range/interval, in which the true value of the mean lies, with 95% probability/chance; <i>idea of</i> the true/AW, mean, lies within, $\pm 2 \times S_M/SE$, with 95% probability/chance; <i>idea of</i> the (calculated) mean is close to the true/actual mean; shows the reliability of the (calculated) mean; (the confidence intervals are small) so data is reliable; (the confidence limits do not overlap) so data is reliable;	1	<i>this must be a clear statement</i> A 95% confident/sure/certain, that the true/actual/population mean lies within this range I ora for 5% I 95% reliable

Question	Answer	Mark	Additional Guidance
1(d)(iii)	<i>t</i>-test ; data has a normal distribution / comparing the <u>means</u> of two samples ;	2	<i>if test not correct allow reason if correct for stated test and <i>t</i>-test e.g. Pearson's linear correlation because gave normal distribution</i> A comparing two means / comparing a pair of means / to see if two means are different A data is continuous / not discrete I continuous variation
1(e)	<i>four from:</i> 1 large number of stomata / 50 stomata (from each epidermal surface) (for each ABA concentration) ; 2 (left for) the same time / left for <u>2 hours</u> ; 3 same age of leaf / young leaves used ; 4 describe how one (stated) environmental condition <u>is</u> controlled ; 5 <i>ref. to</i> how one stated method of measurement has been standardised ; 6 random selection of stomata (to avoid bias) ;	4	I <i>ref. to</i> confidence intervals I large sample size unqualified A 10 stomata from each (epidermal) strip I time unqualified A seedling leaf / leaves just expanded either carbon dioxide-free air or pH by buffer I 'ensure no carbon dioxide in environment' calibrated, eye piece / graticule or same microscope or same magnification
	Total:	19	

--	--	--	--	--

Question	Answer	Mark	Additional Guidance
----------	--------	------	---------------------